

$$\text{MGF} = \mathbb{E}[e^{tX}], \quad \text{CF} = \mathbb{E}[e^{iuX}], \quad \textcircled{t} = \textcircled{iu}$$

$$\phi_x(u) = \mathbb{E}[e^{iuX}] = \int_{-\infty}^{\infty} e^{iuX} f_x(x) dx$$

$$= \mathbb{E}\left[1 + iuX + \frac{1}{2!}(iuX)^2 + \dots + \frac{1}{k!}(iuX)^k + \dots\right]$$

$$= 1 + (iu) \mathbb{E}[X] + \frac{1}{2!}(iu)^2 \mathbb{E}[X^2] + \dots + \frac{1}{k!}(iu)^k \mathbb{E}[X^k] + \dots$$

$$\left. \frac{\partial^k \phi_x(u)}{\partial u^k} \right|_{u=0} = \frac{k!}{i^k} \mathbb{E}[X^k] + \textcircled{X}$$

$$\mathbb{E}[X^k] = \frac{1}{i^k} \cdot \frac{\partial^k \phi_x(u)}{\partial u^k} \leftarrow$$

$$\begin{aligned} \textcircled{1} \phi_{x+y}(u) &= \mathbb{E} \left[e^{iu(x+y)} \right] = \mathbb{E} \left[\underline{e^{iu x}} \cdot \underline{e^{iu y}} \right] \\ &= \mathbb{E} \left[e^{iu x} \right] \cdot \mathbb{E} \left[e^{iu y} \right] \\ &= \phi_x(u) \cdot \phi_y(u) \end{aligned}$$

$$\textcircled{2} \phi_{ax}(u) = \mathbb{E} \left[e^{iu ax} \right] = \phi_x(au)$$

$$\textcircled{3} \underline{x} \sim N(\mu, \sigma^2)$$

$$\phi_x(u) = \mathbb{E} \left[e^{iu x} \right]$$

$$\underline{(iu)x} \sim N(iu\mu, \underline{i^2 u^2 \sigma^2}) = N(\underline{i u \mu}, \underline{-u^2 \sigma^2})$$

$$\phi_x(u) = \mathbb{E} \left[e^{N(\underline{a}, \underline{b^2})} \right] \leftarrow$$

$$\underline{\phi_x(u)} = e^{a + \frac{1}{2} b^2} = \underline{e^{iu\mu - \frac{1}{2} u^2 \sigma^2}}$$

$$\begin{aligned}
 \mathcal{F}(af_1 + bf_2) &= \int_{-\infty}^{\infty} (af_1(x) + bf_2(x)) e^{iux} dx \\
 &= \int_{-\infty}^{\infty} af_1(x) e^{iux} dx + \int_{-\infty}^{\infty} bf_2(x) e^{iux} dx \\
 &= a \cdot \mathcal{F}_1(u) + b \mathcal{F}_2(u)
 \end{aligned}$$

$$\begin{aligned}
 \mathcal{F}(f(x-x_0)) &= \int_{-\infty}^{\infty} f(x-x_0) \cdot e^{iux} dx \\
 \text{with } x-x_0 &= y \\
 &= \int_{-\infty}^{\infty} f(y) e^{iu(y+x_0)} dy \\
 &= \underbrace{(e^{iux_0})}_{\text{constant}} \underbrace{\int_{-\infty}^{\infty} f(y) \cdot e^{iuy} dy}_{\mathcal{F}(u)}
 \end{aligned}$$

① $\delta(x) \longleftrightarrow 1$ (Fourier pair)

$$F(u) = \int_{-\infty}^{\infty} \delta(x) \cdot e^{iux} dx = e^{iu(0)} = 1$$

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} 1 \cdot e^{-iux} du$$

② $F(u) = \int_{-\infty}^{\infty} \delta(x-a) e^{iux} dx = e^{iua}$

③ $f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi \delta(u) \cdot e^{-iux} du = e^{-ix(0)} = 1$

④ $f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi \delta(u-a) e^{-iux} du = e^{-ixa}$

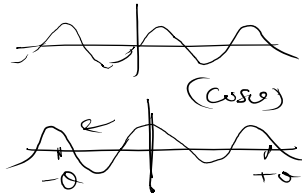
$$F(u) = \pi (\delta(u-a) + \delta(u+a))$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \pi (\delta(u-a) + \delta(u+a)) e^{-iux} du$$



(2nd)

$$= \frac{1}{2} \left[\int_{-\infty}^{\infty} \delta(u-a) \cdot e^{-iux} du + \int_{-\infty}^{\infty} \delta(u+a) e^{-iux} du \right]$$

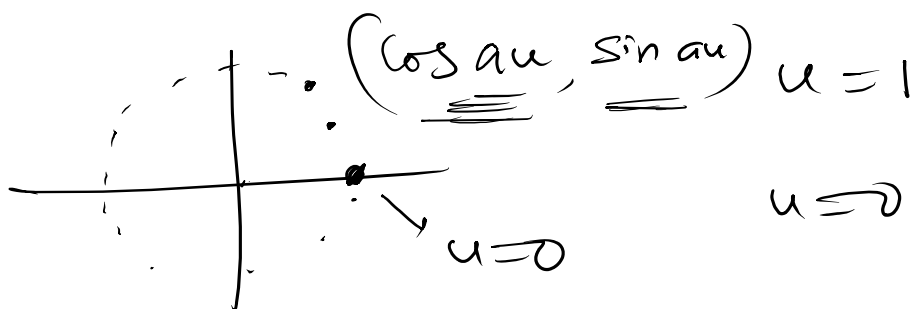


$$= \frac{1}{2} \left[e^{-ixa} + e^{ixa} \right]$$

$$\begin{aligned} e^{i\theta} &= \cos \theta + i \sin \theta \\ e^{-i\theta} &= \cos \theta - i \sin \theta \end{aligned}$$

$$= \frac{1}{2} \left(\cos(ax) - i \sin(ax) + \cos(ax) + i \sin(ax) \right)$$

$$= \cos(ax)$$



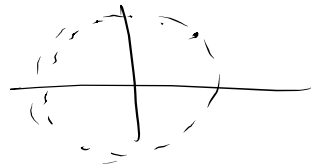
$$\underline{X(T)} = \ln(\underline{S_T}) \sim N(\underbrace{\ln S_0 + (r - \frac{1}{2}\sigma^2)T}_{\mu}, \underbrace{\sigma^2 T}_{\sigma^2})$$

$$\phi_x(u) = e^{i u \mu - \frac{1}{2} \sigma^2 u^2}$$

$$\phi_{X(T)}(u) = \left(e^{\frac{i u [\ln S_0 + (r - \frac{1}{2}\sigma^2)T]}{}} - \frac{1}{2} \sigma^2 T u^2 \right)$$

$$= e^{i u \theta - \frac{1}{2} \sigma^2 u^2 T}$$

$$= \underbrace{\left[e^{-\frac{1}{2} \sigma^2 u^2 T} \right]}_{\text{damping function}} \times \underbrace{\left(e^{i u \theta} \right)}_{\text{oscillating function}} \rightarrow \left(\cos u \theta + i \sin u \theta \right)$$



damping
function